



Active and stable Cu doped NiMgAlO catalysts for upgrading ethanol to *n*-butanol

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ABSTRACT

Upgrading ethanol to *n*-butanol is an attractive way for renewable *n*-butanol production. Herein, Cu was selected to modify NiMgAlO catalysts for improving ethanol conversion and *n*-butanol selectivity. Over the optimized 2%Cu-NiMgAlO catalyst, ethanol conversion and *n*-butanol selectivity were enhanced to 30.0% and 64.2%, respectively, in 200 h time on stream at 523 K. According to physicochemical characterizations and theoretical calculations, the key role of multiple active sites in this reaction was extensively investigated. The plate-like structure of hydrotalcite was maintained over 2%Cu-NiMgAlO catalysts, with an average Ni particle size of ca. 5.4 nm. The presence of Cu species created CuNi alloy sites and Lewis acid-base pairs, and increased hydrogen transfer and condensation reactions, resulting in elevated ethanol conversion and *n*-butanol selectivity. Additionally, CuNi alloy had a strong interaction with CuNiMgAl oxides, forming homogeneous boundary due to their close ionic radius and lattice matching, and afforded the long time stability in the ethanol to *n*-butanol reaction.

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1. Introduction

The excessive consumption of fossil oil has induced serious environmental issues, and stimulates the exploration of “carbon-neutral” chemicals or fuels [1,2]. As one of the most abundant renewable feedstock, ethanol has gained great attention due to its huge production amount and versatile applications. In 2018, the production of ethanol surpassed 30 billion gallons, with an estimated 5% increase for the next several years [3–5]. Nevertheless, the primary ethanol application is greatly confined by the “blend wall”. The maximum percentage of ethanol permitted for use in conventional vehicles is 10 vol% (E10). Although E20 and E85 engines have been used in some flex-fuel vehicles, the production of ethanol is still higher than its consumption in some specific areas due to the rapid development of fermentation technologies and newly commercialized coal (syngas) based ethanol. Besides being used as a fuel additive, ethanol is also an attractive platform chemical for upgrading, such as dehydrogenation to acetaldehyde,

condensation to propylene, *n*-butanol, 1,3-butadiene, isobutene and aromatics [6–14]. Among these catalytic pathways, conversion of ethanol to *n*-butanol is highly attractive due to the versatile applications of *n*-butanol and the insufficient production of renewable *n*-butanol from the ABE (Acetone-Butanol-Ethanol) process [15–17]. In past decades, catalytic conversion of ethanol to *n*-butanol has been hotly investigated and achieved remarkable success. Several kinds of catalysts, such as homogeneous catalysts, metal oxides, binary catalysts and metal-enhanced oxide catalysts, have been developed for this reaction [18–23]. From the viewpoint of catalyst activity and reusability, metal-enhanced oxide catalysts have great advantages due to the synergistic effect between metals and oxides in coupling hydrogen transfer and condensation reactions for this transformation [6,24].

Layered double hydroxides (LDHs) are a series of two-dimensional anionic clays, consisting of positively charged brucite-like host layers, water, inorganic or organic charge-compensating anions confined in the interlayer gallery [25–27]. In this specific material, both cationic clays and interlayer anions are exchangeable, providing an interesting catalyst or support for heterogeneous catalysis. Moreover, thanks to the topological structural transformation of LDHs, highly dispersed mixed metal oxides

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with adjustable acidic and basic properties have been widely synthesized, and applied in ethanol to *n*-butanol reactions. Cosimo et al. synthesized different Mg₂AlO_x catalysts from LDHs for ethanol coupling, and disclosed the possible reaction pathway to *n*-butanol [28]. Afterwards, metal-enhanced mixed oxide catalysts were synthesized for ethanol conversions. The presence of transition metal sites promoted the hydrogen transfer ability of metal oxides, and correspondingly elevated the ethanol conversion with considerable *n*-butanol selectivity [29–34]. In our previous research, NiMgAlO catalysts were synthesized for ethanol conversion, and obtained 55.2% selectivity to *n*-butanol and 85% selectivity to C₄–C₈ alcohols at 18.7% ethanol conversion under the condition of 523 K and a weight hourly space velocity (WHSV) of 3.2 h^{−1} [35]. Although metal, strong base and weak acid sites cooperated in this cascade reaction, both ethanol conversion and *n*-butanol selectivity are yet to be improved. Cu-based porous metal oxides were widely developed for this reaction to enhance ethanol conversion or *n*-butanol selectivity. For instance, Barrett et al. used copper-doped porous metal oxide catalysts with ppm quantities of soluble chloride sources to elevate the ethanol conversion to ca. 60% with 55% *n*-butanol selectivity in a batch reactor at 593 K. However, the presence of Cl[−] still caused the separation and recycle issues, and to balance ethanol conversion and *n*-butanol selectivity is still a long-term goal. More important, the underlying reason for the promotion effect of Cu based catalysts is yet to be disclosed [9,30,31].

For the preparation of metal-enhanced oxide catalysts, transition metals are generally introduced as divalent and/or trivalent metal cations during the LDHs synthesis stage. Additionally, some metals are preferably brought in by metal ion adsorption manners, providing an optional way for metal-based bifunctional catalysts synthesis, and an elegant strategy to disclose the versatile role of metals in this reaction. Herein, using the NiMgAlO catalyst as a precursor, different metals were introduced by the metal ion adsorption method to improve ethanol conversion and *n*-butanol selectivity. Over the optimized 2%Cu-NiMgAlO catalyst, effects of reaction temperature and WHSV on this ethanol to *n*-butanol reaction were investigated, and the stability of catalysts before and after doping Cu was evaluated. Thereafter, the physical and chemical structures of 2%Cu-NiMgAlO catalysts were characterized to unveil the key role of atomically dispersed Cu in promoting the ethanol to *n*-butanol reaction.

2. Experimental

2.1. Catalysts preparation

2.1.1. Preparation of NiMgAlO and CuMgAlO LDHs

All LDHs were prepared using a co-precipitation method. Taking the NiMgAlO LDHs as an example, 50 mL aqueous solution containing 0.02 mol Ni(NO₃)₂·6H₂O, 0.08 mol Mg(NO₃)₂·6H₂O, and 0.02 mol Al(NO₃)₃·9H₂O was dropwise added to a fiercely stirred 50 mL solution containing 0.08 mol Na₂CO₃ and 0.12 mol NaOH. The pH of the obtained mixture was adjusted to 10 by a 3 mol/L NaOH solution. The mixture was kept at 338 K for 18 h, and a hydrotalcite structure of NiMgAlO precursor was formed. The hydrotalcite was filtered, rinsed with 500 mL deionized water, dried at 393 K for 12 h, and finally got the LDHs of NiMgAlO with a Ni:Mg:Al molar ratio of 1:4:1.

2.1.2. Preparation of NiMgAlO and CuMgAlO catalysts

The obtained LDHs of NiMgAlO and CuMgAlO were calcined at 873 K in the air for 2 h, reduced at 873 K in hydrogen for 1 h, and obtained NiMgAlO and CuMgAlO catalysts.

2.1.3. Preparation of Cu-NiMgAlO catalysts

Cu-NiMgAlO catalysts were prepared by the metal ion adsorption method [36]. In a typical run, 20 mL aqueous solution containing a certain amount of Cu(NO₃)₂·3H₂O was dropwise added to 50 mL fiercely stirred solution containing 0.5 g LDHs of NiMgAlO, and further stirred at 338 K for 1 h. Then, the pH of obtained mixture was adjusted to 10 by 0.2 mol/L NaOH solution, and maintained fiercely stirring for 4 h. After the filtration process, the resulted slurry was washed with 100 mL deionized water, and dried at 393 K for 12 h. Finally, the obtained powder was calcinated at 873 K for 2 h, reduced at 873 K in hydrogen for 1 h, and yielded Cu-NiMgAlO catalysts. The number before Cu was the theoretical amount of Cu loading calculated based on the amount of NiMgAlO-hydrotalcites (HT), e.g., 1%Cu-NiMgAlO represents the theoretical Cu loading in the catalyst was 1%. Scheme 1.

2.2. Characterizations

X-ray diffraction (XRD) patterns of different NiMgAlO precursors and catalysts were performed on a PANalytical X'Pert-Pro powder X-ray diffractometer using Cu K α monochromated radiation (λ = 0.15418 nm) at 40 kV and 40 mA. Before analysis, samples were ground to powder larger than 100 meshes, and loaded onto a glass slide. The patterns were recorded in 2 θ range 10–80 ° at a scan speed of 5 °/min. PANalytical X'Pert HighScore Plus software was used for qualitative analysis.

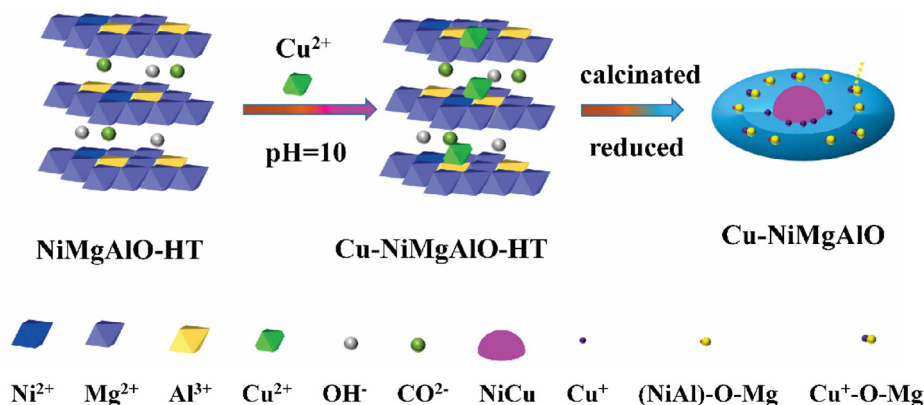
The pore structure of NiMgAlO based catalysts was measured using a Micromeritics ASAP 2020 system with nitrogen as the adsorbent at 77 K. Prior to nitrogen adsorption, about 0.1 g catalyst was out-gassed at 393 K for 3 h, and then at 523 K for 3 h. The specific surface area was determined by the BET method, pore volume and average pore diameter were calculated by the BJH method.

High-angle annular dark field scanning transmission electron microscopy (HAADF-STEM) and energy dispersive spectroscopy (EDS) elemental mapping were conducted on a JEM-2100F microscope. Before analysis, the sample was ground and dispersed in ethanol under ultrasonic conditions for 10 min. The dispersed sample was then mounted onto carbon-coated Mo grids followed by slow evaporation of ethanol at room temperature.

X-ray photoelectron spectroscopy (XPS) of different catalysts was recorded on a ThermoFisher spectrometer using Al K α ($h\nu$ = 1486.6 eV) X-ray source with a pass energy of 30 eV, and a base pressure of 5 × 10^{−9} mbar in analysis chamber. The spectra before and after 10 s Ar etching were collected. Binding energy (BE) values were referenced to the binding energy of the C 1s core level at 284.6 eV.

Pseudo X-ray absorption spectroscopy (XAS) was performed at the BL 14 W1 of Shanghai Synchrotron Radiation Facility, Shanghai Institute of Applied Physics, China, and the X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) of catalysts were analyzed accordingly. The sample was pretreated in a H₂ flow (60 mL/min) at 873 K for 2 h, followed by sealing with Capton film in a glove box without exposure to air. The spectra were collected at room temperature in the transition mode.

Temperature-programmed reduction (TPR) and temperature-programmed desorption (TPD) experiments were carried out on a Micromeritics AutoChem II 2920 automated system equipped with a thermal conductivity detector (TCD). For TPR experiments, the sample was pretreated with Ar at 393 K for 1 h to remove the adsorbed water. The gas was then switched to 10 vol% H₂/Ar, and the temperature was increased to 1073 K at a rate of 10 K/min. For H₂, CO, and NH₃-TPD experiments, the catalyst was first pretreated and then saturated with pulsed H₂, CO, and NH₃ through the built-in gas sampling loop, and finally the adsorbate was des-



Scheme 1. Procedures for the synthesis of Cu-NiMgAlO catalysts.

orbed at different temperatures. Taking the H_2 -TPD experiment as an example, the catalyst was first reduced in hydrogen at 873 K for 1 h, and then flushed with Ar at 883 K for 0.5 h to remove the surface adsorbed hydrogen. After being cooled down to 323 K, the sample was titrated with pulsed H_2 from the built-in gas sampling loop until saturation. Finally, the sample was flushed with Ar, and the TPD signal was collected from 323 to 973 K at a rate of 10 K/min. The H_2 , CO and NH_3 adsorption amounts were calculated from pulse chemisorption peaks.

In situ Diffuse Reflectance Infrared Fourier Transform Spectroscopy (DRIFTS) experiments were conducted on a Bruker Tensor 27 FT-IR spectrometer at a resolution of 4 cm^{-1} between 4000 cm^{-1} and 600 cm^{-1} . About 20 mg catalyst was loaded into an *in situ* cell and reduced at 773 K for 1 h in 10% H_2 /Ar at a rate of 50 mL/min. After being cooled to 323 K, a background spectrum was collected, following which probe molecules of CO were dosed (10 μL 10% CO/Ar) and IR spectra were recorded every 60 s until saturated adsorption was achieved. The system was then purged with Ar to remove physically adsorbed CO and IR spectra were recorded. Finally, the sample was dosed with ethanol for 20 times, and treated at different temperatures for IR spectra collection.

Inductively coupled plasma-atomic emission spectrometry (ICP-AES) analyses were conducted to determine the metal content in catalysts. Before the ICP-AES analysis, samples were dissolved in aqua regia (3 mL HCl and 1 mL HNO_3 , diluted to 10 mL, 50 mL or 100 mL according to the concentration of metal ions).

2.3. Computational details

All calculations were carried out over the Gaussian 09 software. The PBE0 functional was adopted for all calculations in combination with the D3BJ dispersion correction. For geometry optimization and frequency calculations, the 6-31G(d) for basis set and Hay's Gaussian type basis set for Cu^6 were employed. The singlet point energy calculations were performed with the 6-311 + G(d, p) basis set. The Gibbs free energy was obtained by adding the single point energy with the thermal correction by frequency calculation [37].

2.4. Catalytic reaction

For screening different metal-doped NiMgAlO catalysts, reactions were carried out in a 50 mL Hastelloy autoclave. In a typical run, 0.2 g catalyst and 20 mL ethanol (analytical ethanol, 99.5 wt%) were charged into the autoclave. The reactor was then flushed with nitrogen three times and filled with 0.1 MPa nitrogen. Finally, the mixture was stirred at 700 r/min and heated to 503 K for 24 h.

Other experiments were performed in a high-pressure fixed-bed reactor. In a typical experiment, 1.4 g catalyst was placed at the center of a steel reactor with 50–70 meshes quartz on both sides of the reactor. After *in situ* reduced at 873 K with pure hydrogen for 1 h, the reactor was cooled to the reaction temperature. Then, the gas was switched to N_2 at 60 mL/min and pressed to 3 MPa. Finally, analytically pure ethanol was pumped into the reactor using a high-performance liquid chromatography pump. The liquid product was collected in a gas-liquid separator. Gaseous products were analyzed using an Agilent gas chromatography (GC) system equipped with a TDX-01 column. Liquid products were analyzed using an Agilent GC system equipped with a Varian CP-WAX58 (FFAP) CB capillary column.

Product yields, ethanol conversions and selectivities were calculated by the following equations.

$$\text{Yield (\%)} = \frac{\text{Moles of carbon in the product}}{\text{Moles of carbon in ethanol fed into the reactor}} \times 100$$

$$\text{Conversion (\%)} = \frac{\text{Moles of carbon in ethanol reacted}}{\text{Moles of carbon in ethanol fed into the reactor}} \times 100$$

$$\text{Selectivity (\%)} = \frac{\text{Moles of carbon in the product}}{\text{Moles of carbon in ethanol reacted}} \times 100$$

where, moles of ethanol reacted were calculated based on moles of carbon in all liquid and gaseous products.

3. Results and discussion

3.1. Catalytic conversion of ethanol over different catalysts

3.1.1. Optimize metal-doped NiMgAlO catalysts for ethanol conversion

Different metals were introduced into the hydrotalcite of NiMgAlO by the metal ion adsorption method, and used for ethanol conversions. As shown in Fig. 1(a), over the benchmark catalyst of NiMgAlO, ethanol conversion and *n*-butanol selectivity were ca. 18% and 55.4%, respectively, at 503 K for 24 h. After introducing metals of Fe, Co, Cu, Pt, Pd, Rh and Ir, distinct performances were observed for the conversion of ethanol to *n*-butanol. The introduction of Fe and Rh slightly decreased the ethanol conversion due to their insufficient activity in dehydrogenation reactions, and the presence of Fe, Pt, Rh and Ir reduced the *n*-butanol selectivity to 39–48% because of their high activity in C–C bond cleavages [38,39]. In contrast, Co, Cu and Pd improved both ethanol conversion and *n*-butanol selectivity. Specifically, over the 1%Cu-NiMgAlO

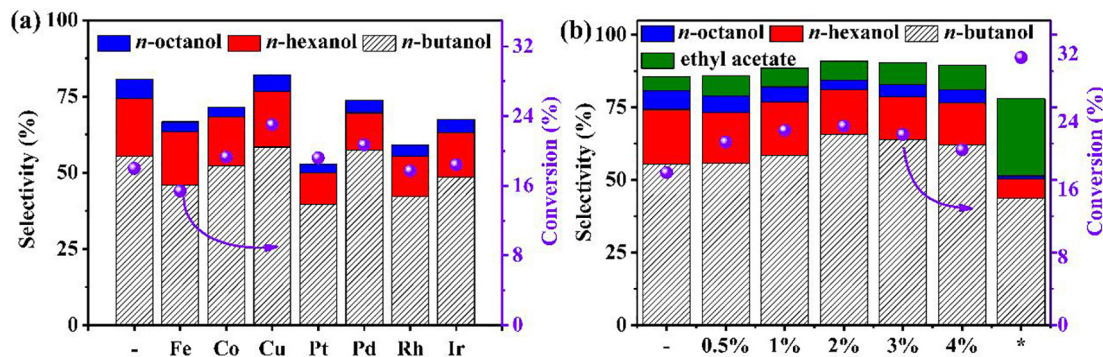


Fig. 1. Catalytic conversion of ethanol over (a) different metal-doped NiMgAlO catalysts and (b) different loading of Cu doped NiMgAlO catalysts (Reaction conditions: 0.2 g catalyst, 20 mL ethanol, 503 K, 24 h, 50 mL autoclave; *Indicates the CuMgAlO catalyst).

catalyst, ethanol conversion and *n*-butanol selectivity reached 23.0% and 58.4%, respectively, demonstrating the unique role of Cu in promoting the ethanol to *n*-butanol reaction [11,40]. Effects of Cu content on ethanol conversion were then investigated and shown in Fig. 1(b). Ethanol conversion and *n*-butanol selectivity first increased and then decreased slightly, reaching the maximum values of 23.5% and 65.7%, respectively, over the 2%Cu-NiMgAlO catalyst. For the CuMgAlO catalyst with a higher Cu content, the ethanol conversion increased to 31.6% at the expense of *n*-butanol selectivity (43.6%). The key role of Cu species in changing ethanol conversion and *n*-butanol selectivity should be attributed to different structures and chemical properties between NiMgAlO and Cu-NiMgAlO catalysts [41–43].

3.1.2. Effects of reaction conditions on the ethanol to *n*-butanol reaction

After screening metals on NiMgAlO, the optimized 2%Cu-NiMgAlO catalyst was loaded into the fixed-bed reactor for investigation. The dependence of temperature on ethanol conversion and product selectivity was first studied in Fig. 2(a), and their corresponding HPLC profiles were shown in Fig. S1. At 373 K, the ethanol conversion was merely 0.9%, with acetaldehyde and ethyl acetate as the main product (36.2%). Increasing temperature to 573 K linearly elevated ethanol conversion to 37.6%. Differently, the *n*-butanol selectivity increased to 61.8% at 473 K, then fluctuated to 59.3% at 548 K, and finally decreased significantly to 31.7% at 573 K. At low temperatures of 373–448 K, the condensation reaction was inhibited and resulted in large amounts of acetaldehyde and ethyl acetate via dehydrogenation and Tishchenko reactions. After increasing temperatures to 473–548 K, the condensation reaction was promoted, and *n*-C₄–C₈ alcohols selectivity was raised to ca. 80%. Nevertheless, at the high temperature of 573 K, the decomposition of ethanol and *n*-C₄–C₈ alcohols happened, and caused large amounts of gaseous products (16.8%) [44]. Thus, the *n*-butanol selectivity demonstrated a volcano shape as a function of reaction temperatures.

The influence of WHSV on this reaction was then investigated in Fig. 2(b). At a low WHSV of 0.3 h^{−1}, the ethanol conversion was 42.1% with large amounts of gaseous products, resulting in a low carbon balance, similar to the reaction result at 573 K. In contrast, at the WHSV range of 0.6–8.4 h^{−1}, the ethanol conversion gradually decreased to 20.0% as increasing the WHSV to 8.4 h^{−1}, but the *n*-butanol selectivity maintained >60% with 11% *n*-hexanol selectivity in the whole range.

3.1.3. Stability of NiMgAlO based catalysts

The stability of the 2%Cu-NiMgAlO catalyst was evaluated with the comparison of NiMgAlO counterpart. As shown in Fig. 3(a), in the 100 h time on stream, the ethanol conversion was ca. 20% with

51%–58% selectivity to *n*-butanol, ca. 15% selectivity to *n*-hexanol and 3% selectivity to *n*-octanol over the NiMgAlO catalyst. Afterwards, both ethanol conversion and product selectivity decreased in the 100–120 h time on stream. In contrast, the 2%Cu doped NiMgAlO catalyst demonstrated improved activity and stability in Fig. 3(b). Specifically, the ethanol conversion and *n*-butanol selectivity increased to 30% and 62%–64%, respectively, with a slight decrease in *n*-hexanol and *n*-octanol selectivities. Moreover, this activity was maintained for more than 200 h time on stream without any deactivation, indicating the high stability of the Cu-NiMgAlO catalyst for ethanol conversion.

3.2. Catalyst characterization

3.2.1. XRD, pore structure and TEM of different catalysts

XRD patterns of NiMgAlO, CuMgAlO and 2%Cu-NiMgAlO precursors and corresponding catalysts were shown in Fig. 4(a). Hydrotalcites of NiMgAlO, CuMgAlO and 2%Cu-NiMgAlO exhibited typical diffraction patterns at 2θ angles of 11.4°, 22.7°, 34.4°, 60.3° and 62.0°, corresponding to (003), (006), (012), (110) and (113) planes of LDHs (PDF #00-022-0700), respectively [45]. Upon calcination at 873 K, the structure of LDHs was collapsed and converted to their corresponding atomically dispersed mixed oxides. After reduction at 873 K, major diffraction peaks at 2θ values of 36.4°, 43° and 62.3° were observed, indexed to (111), (200) and (220) planes of MgO (PDF #01-075-1525). Additionally, a weak peak at 51.8° was also detected, which should be attributed to the metallic Ni phase (PDF # 01-070-0989). For Cu-NiMgAlO catalysts with different Cu loadings in Fig. 4(b), no peak of Cu species was observed in the whole range of Cu loadings, indicating the high dispersion of Cu species over NiMgAlO catalysts. The typical peak of the metallic Ni phase maintained at 51.8° at low Cu loadings of 0–2%, and shifted to 50.9° and 50.3° at high Cu loadings of 4% and 8%, respectively, indicating the formation of CuNi alloys at high Cu dopings [46–48]. The 2%Cu-NiMgAlO catalysts before and after reaction were characterized by XRD patterns. As shown in Fig. S2, catalysts before and after reaction demonstrated identical peaks, confirming the high stability of Cu-NiMgAlO based catalysts in ethanol conversions.

Metal contents and pore structure of catalysts were analyzed and shown in Table 1. For the NiMgAlO catalyst, Ni, Mg and Al contents were 20.1%, 35.6% and 10.9%, respectively, close to their theoretical amounts. After doping with Cu, percentages of Ni, Mg and Al were fluctuated around 20%, 31% and 9%, respectively. The contents of Cu were higher than their theoretical amounts calculated based on the weight of NiMgAlO-HT, which should be attributed to the removal of water and CO₂ during the calcination process. The specific surface area of NiMgAlO was 176 m²/g, which gradually increased to 231 m²/g at 2% Cu loading and then decreased

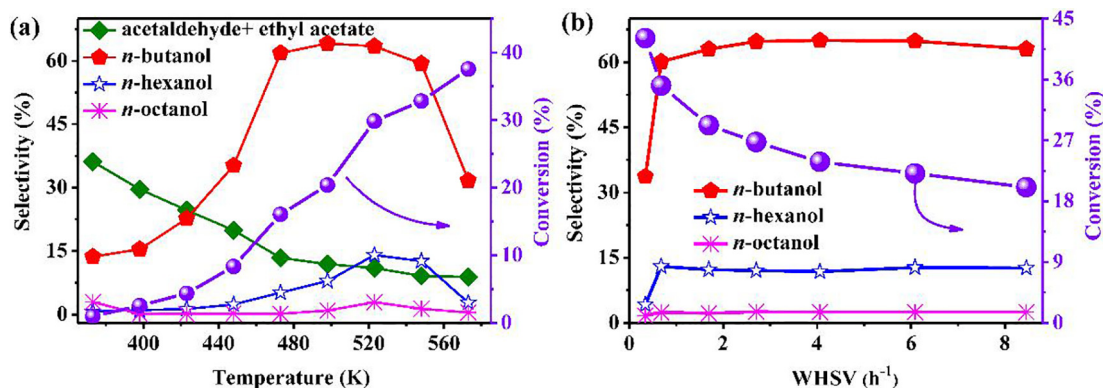


Fig. 2. Catalytic conversion of ethanol over 2%Cu-NiMgAlO catalysts at different (a) temperatures (WHSV = 1.69 h⁻¹) and (b) WHSV (523 K) (Reaction conditions: 3 MPa 60 mL/min N₂, 99.5% ethanol).

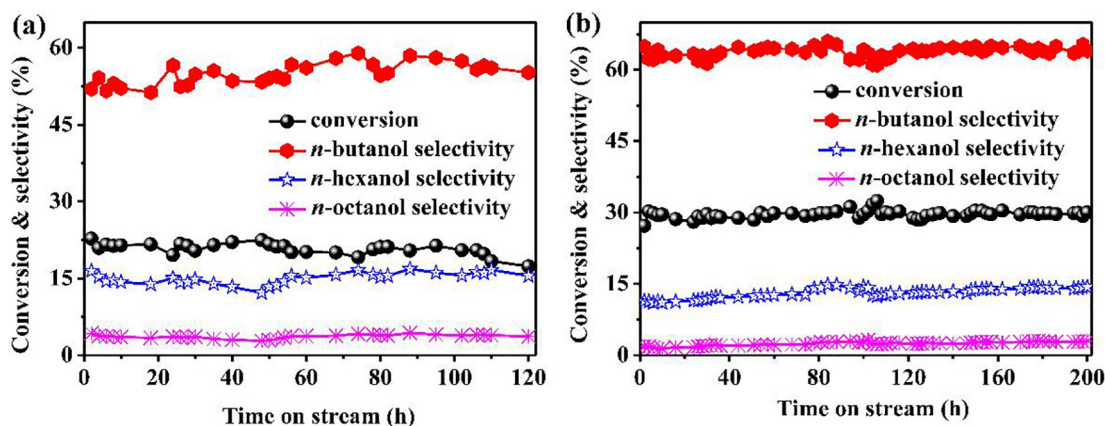


Fig. 3. Stability of (a) NiMgAlO and (b) 2%Cu-NiMgAlO catalysts (Reaction conditions: 3 MPa 60 mL/min N₂, 99.5% ethanol, 523 K, WHSV = 1.69 h⁻¹).

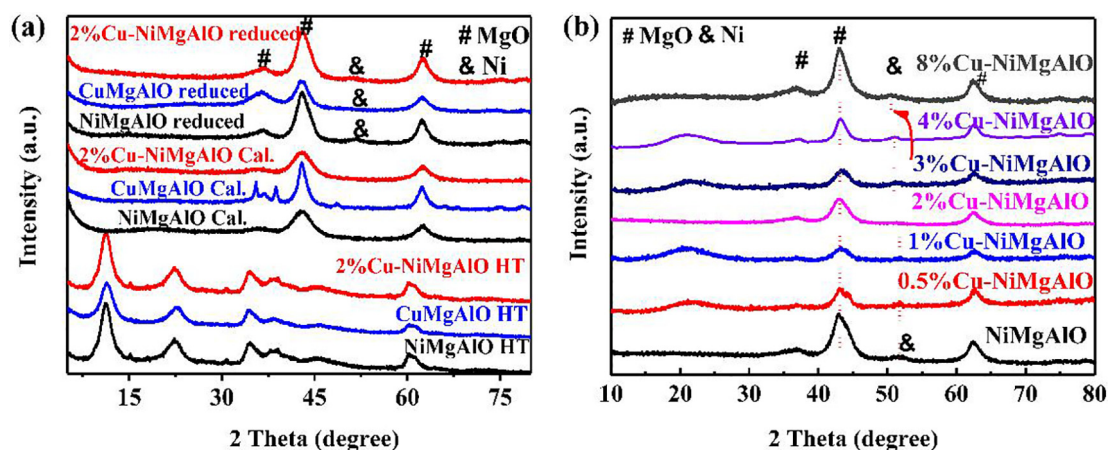


Fig. 4. XRD patterns of (a) NiMgAlO, CuMgAlO and 2%Cu-NiMgAlO precursors and corresponding catalysts and (b) Cu-NiMgAlO catalysts at different Cu loadings.

to 169 m²/g at 4% Cu doping. During the catalyst preparation process, NiMgAlO-HT strongly adsorbed the introduced Cu species to interlayers, which enlarged the specific surface area of NiMgAlO based catalysts after calcination and reduction processes [49]. In contrast, the excessive Cu loading (4% and 8% Cu dopings) induced the formation of bulk CuNi alloys, and slightly decreased the specific surface area of Cu-NiMgAlO catalysts.

HRTEM images of NiMgAlO, 2%Cu-NiMgAlO and 4%Cu-NiMgAlO catalysts were obtained and shown in Fig. 5. For the NiMgAlO cat-

alyst (Fig. 5a), black dots with an average diameter of 5.2 nm were observed in the plate-like structure, which should be attributed to Ni particles embedded in hydrotalcite precursor derived mixed oxides. In the enlarged image (Fig. 5a'), the lattice spacing of 0.216 nm was detected in typical areas, which should be assigned to the (220) plane of Al modified MgO, consistent with the XRD result (PDF #01-075-1525) in Fig. 4. Additionally, lattice spacings of 0.207 nm attributed to the (111) plane of metallic Ni particles were observed. After introducing 2% Cu, the plate-like morphology

Table 1
Metal contents and pore structure of NiMgAlO, CuMgAlO and Cu-NiMgAlO based catalysts^a.

Catalyst	Content (%) ^b				S_{BET} ($\text{m}^2 \text{g}^{-1}$)	V (cc g^{-1})	D_p (nm)
	Cu	Ni	Mg	Al			
NiMgAlO	–	20.1	35.6	10.9	176	0.42	3.9
CuMgAlO	22.3	–	32.1	8.9	104	0.47	7.8
0.5%Cu-NiMgAlO	0.8	22.4	32.3	9.8	183	0.75	6.1
1%Cu-NiMgAlO	1.4	21.5	32.5	8.9	196	0.70	3.2
2%Cu-NiMgAlO	3.2	20.1	32.2	8.7	213	0.63	2.7
3%Cu-NiMgAlO	5.6	20.2	31.4	8.7	198	0.55	3.2
4%Cu-NiMgAlO	8.3	20.4	30.5	8.5	169	0.59	3.2
8%Cu-NiMgAlO	15.8	18.4	28.1	7.6	154	0.48	3.6
2%Cu-NiMgAlO used	3.4	20.2	29.7	9.3	162	0.53	5.5

^a S_{BET} , V and D_p are abbreviations for the specific surface area calculated by the BET method, the volume determined according to the BJH method, and the average pore diameter, respectively.

^b The actual Ni, Cu, Mg, Al contents were measured by the ICP-AES.

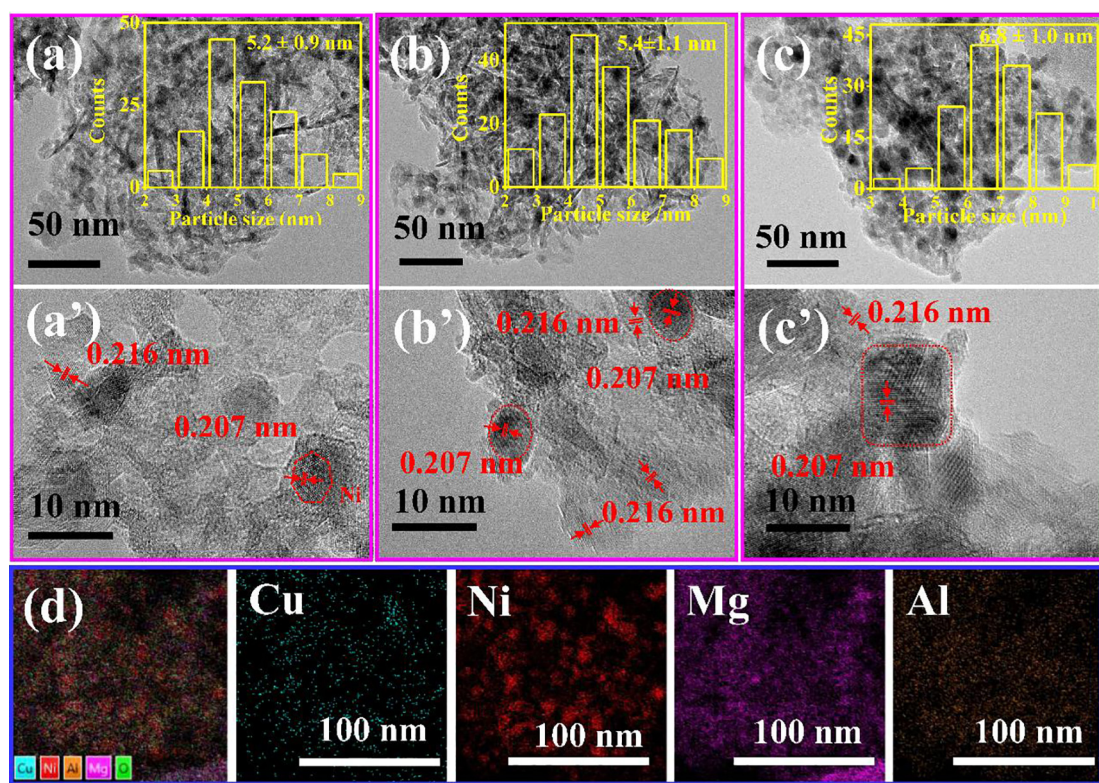


Fig. 5. HRTEM images of (a and a') NiMgAlO, (b and b') 2%Cu-NiMgAlO and (c and c') 4%Cu-NiMgAlO catalysts and their particle size distributions (inset in a–c), and (d) EDS elemental mappings of Cu, Ni, Mg and Al for the 2%Cu-NiMgAlO catalyst.

was maintained with a slight increase of the average particles size to 5.4 nm, as depicted in Fig. 5(b). In the enlarged image (Fig. 5b'), typical lattice spacings attributed to Al modified MgO, and Ni(Cu) were also visualized. Besides, the Al modified MgO and Ni(Cu) were closely interacted, forming a homogeneous boundary due to their close ionic radius and lattice matching between (220) plane of MgO and (111) plane of Ni(Cu) (MgO: 0.2099 nm vs Ni(Cu) 0.202 nm). In contrast, the presence of 4% Cu greatly changed the morphology of the NiMgAlO matrix and the particle size of Ni (Cu). As shown in Fig. 5(c), the plate-like morphology disappeared, and the average particle size of Ni(Cu) increased to ca. 6.8 nm, much higher than that of NiMgAlO and 2%NiMgAlO catalysts. The enlarged image (Fig. 5c') showed the lattice matching of Al modified MgO and Ni(Cu). Nevertheless, some lattice spacings of Ni were modified by Cu species due to CuNi alloys formation, as evidenced by the EDS mapping of the 4%Cu-NiMgAlO catalyst in

Fig. S3. The undetected changes in lattice spacings should be attributed to the close (111) plane of metallic Ni and Cu (Ni: 0.202 nm vs Cu 0.2087 nm). The optimized 2%Cu-NiMgAlO catalyst was also characterized by the EDS mapping in Fig. 5(d). Ni particles were clearly observed, similar to that of TEM images. Differently, Cu species were nearly atomically dispersed and could not be detected by TEM. According to the result of 4%Cu-NiMgAlO catalyst, we deduced that most Cu species on the 2%Cu-NiMgAlO catalyst strongly connected with Ni or MgO, forming dilute CuNi alloy and Cu modified MgO species that had a negligible effect on the size of Ni particles. The Cu-NiMgAlO catalyst before and after the reaction was characterized by HRTEM. As shown in Fig. 5(b–b') and Fig. S4, the used Cu-NiMgAlO catalyst demonstrated similar particle size distribution and identical lattice spacings compared with the fresh counterpart, indicating the high stability of Cu-NiMgAlO catalysts in ethanol conversions. In all, the strong

interaction between CuNi alloy and CuNiMgAl oxides formed a homogeneous boundary due to their close ionic radius and lattice matching, which inhibited the aggregation of metal particles, and maintained its high stability in this reaction.

3.2.2. XPS and XAS of NiMgAlO and Cu-NiMgAlO catalysts

XPS spectra of NiMgAlO and Cu-NiMgAlO catalysts before and after Ar etching were collected. In contrast to notable oxygen species on the surface of un-pretreated samples, Ar etched counterparts preferably exposed metallic Ni/Cu species, which reflected the original chemical state of metallic species in fresh catalysts (Fig. S5). Thus, Ar etched catalysts were used for analysis, as shown in Fig. 6.

In Fig. 6(a), three peaks at 861.4 eV, 855.3 eV and 851.9 eV were detected for NiMgAlO and Cu-NiMgAlO catalysts, and should be ascribed to Ni satellite peak, $\text{Ni}^{2+} 2p_{3/2}$ and $\text{Ni}^0 2p_{3/2}$, respectively. The appearance of Ni^{2+} species on the surface of the reduced catalyst indicated that some Ni species had a strong interaction with $\text{MgO}(\text{Al}_2\text{O}_3)$, and were hardly being reduced into metallic Ni [50]. Due to the key role of Ni in this reaction, the percentage of metallic Ni was quantified through normalizing the area of the corresponding featured peaks. The percentage of Ni^0 over the NiMgAlO catalyst was 13.9% with 86.1% Ni reserved as Ni^{2+} . After the introduction of Cu, the percentage of Ni^0 increased to 21.7% over the 2%Cu-NiMgAlO catalyst. Simultaneously, the presence of Cu changed the surface oxygen species of the NiMgAlO matrix. As shown in Fig. 6(b), O 1s core level depicted two distinct peaks at 531.4 and 529.7 eV, which should be assigned to strong base sites of lattice oxygen (O^{2-}) and weak base sites of hydroxyl oxygen, respectively [51,52]. For the NiMgAlO catalyst, the percentage of O^{2-} was 38.5%, which was gradually increased to 53.6% with the presence of 0.5%Cu, and 59.7% when 2%Cu was added. Correlating with the elevated percentage of metallic Ni, the increased percentage of lattice oxygen species should be attributed to the dehydration of hydroxyl oxygen in the presence of Cu(Ni) species, which increased the strength of base sites for condensation reactions.

The XPS spectra of Cu 2p for 0.5%Cu-NiMgAlO, 2%Cu-NiMgAlO and CuMgAlO catalysts were shown in Fig. 6(c). Two distinct peaks at 931.7 eV and 951.5 eV were observed and should be ascribed to $\text{Cu} 2p_{3/2}$ and $\text{Cu} 2p_{1/2}$, respectively. The Cu $2p_{3/2}$ at 931.7 eV was then deconvoluted into two separated peaks, i.e., 931.6 eV and 933.1 eV, and should be assigned to Cu^+/Cu^0 and Cu^{2+} species [53]. As increasing Cu contents, the percentage of Cu^{2+} gradually decreased, and reached 69.9% over the optimized 2%Cu-NiMgAlO catalyst, indicating the synergistic effect present between Cu and Ni for reduction to yield Cu^+ and metallic Cu in CuNi alloys at high Cu loadings. The distribution of Cu^0 and Cu^+ species was distinguished by the Cu LMM spectra in Fig. 6(d). Asymmetry and broad Auger kinetic energy peaks were observed and deconvoluted into two symmetrical peaks centered at 912.0 eV and 917.8 eV, corresponding to Cu^+ and Cu^0 species, respectively. The percentages of Cu^+ and Cu^0 species were quantified and shown in Table S1. At 0.5% Cu doping, most Cu species were reserved as Cu^0 with 14.8% Cu^+ species. Increasing the Cu loading to 2%, the percentage of Cu^+ species increased to 35.8%, very similar to that of CuMgAlO catalysts. In all, over the optimized 2%Cu-NiMgAlO catalyst, ca. 70% Cu species (Cu^{2+}) had a strong interaction with Ni^{2+} or Mg^{2+} , reluctant to be reduced under present conditions, while ca. 20% Cu species (Cu^0) were nearly atomically dispersed in metallic Ni to CuNi alloy, and ca.10% Cu species (Cu^+) existed in the interface between Cu^0 and $\text{Cu}^{2+}/\text{Mg}^{2+}$. These Cu species provide a useful “pool” for hydrogen adsorption and desorption, and facilitate the hydrogen transfer in this ethanol to *n*-butanol reaction.

X-ray absorption fine structure (XAFS) spectroscopy at Cu K-edge was implemented to study the local structure and coordination environment of the Cu-NiMgAlO catalyst. Fig. 7(a) illustrated

the normalized XANES spectra for the Cu-NiMgAlO catalyst, as well as Cu standard samples. The absorption edges of Cu foil, Cu_2O and CuO were 8980.1 eV, 8980.9 eV and 8984.5 eV, representing the characteristic fingerprint peak of Cu^0 , Cu^+ and Cu^{2+} , respectively. The Cu-NiMgAlO catalyst demonstrated a broad absorption edge at 8980–8985 eV, suggesting the co-existed of Cu^0 , Cu^+ and Cu^{2+} -species [11]. By fitting this XANES spectrum, the amount of Cu^0 , Cu^+ and Cu^{2+} species was estimated to be 39.8%, 16.6% and 43.6%, respectively. The ratios of $\text{Cu}^+/(\text{Cu}^+ + \text{Cu}^0)$ and $(\text{Cu}^+ + \text{Cu}^{2+}) / \text{Cu}^0$ were 0.29 and 1.51, respectively, lower than the XPS result in Table S1 (0.36 and 4.17), which should be attributed to the *ex-situ* characterization of XPS caused the oxidation of Cu species. The k^3 -weighted Fourier transform spectra (*R* space) of standard samples and the Cu-NiMgAlO catalyst were shown in Fig. 7(b). The Cu-NiMgAlO catalyst displayed one strong peak at 2.14 Å, slightly lower than the Cu foil (2.24 Å), indicating the formation of CuNi alloys, rather than bulk Cu particles. Moreover, both the Cu–O peak at 2.76 Å and Cu–O peak at 1.59 Å were shifted compared to the benchmark peaks of CuO and Cu_2O , indicating the strong interaction between Ni, MgO and $\text{Cu}^+/\text{Cu}^{2+}$ species [54].

3.2.3. Chemisorption properties of NiMgAlO and Cu-NiMgAlO catalysts

Conversion of ethanol to *n*-butanol is highly related to the hydrogen transfer ability of catalysts. Thus, H_2 -TPR and H_2 -TPD experiments were conducted to evaluate NiMgAlO and 2%Cu-NiMgAlO catalysts. As shown in Fig. 8(a), two peaks at 727 K and 974 K were observed for the NiMgAlO catalyst, which should be attributed to the reduction of highly dispersed superficial Ni and partially embedded Ni that had a strong interaction with Mg, Al oxides, respectively. After introducing Cu species, the first peak shifted to a lower temperature of 534 K, which should be attributed to the formation of CuNi alloys between Cu and Ni oxides during the reduction process [55], agreeing with previous XRD, TEM and XAFS characterizations. The other peak at ca. 900–950 K assigned to the reduction of partially embedded Ni did not change significantly, implying that the post introduced Cu located on the surface of NiMgAlO catalysts and was highly accessible to reactant molecules. The H_2 -TPD in Fig. 8(b) was further employed to reveal the hydrogen adsorption ability of NiMgAlO and 2%Cu-NiMgAlO catalysts. For the NiMgAlO catalyst, two peaks at 619 K and 981 K were observed, which were attributed to hydrogen adsorbed on metallic Ni centers and hydrogen transferred from metallic Ni sites to Mg, Al oxide supports, respectively. In contrast, a new peak at 376 K was detected for the 2%Cu-NiMgAlO catalyst, indicating the elevated metal active sites after the introduction of Cu. The hydrogen adsorption amount was calculated to be 71.5 $\mu\text{mol/g}$ for the 2%Cu-NiMgAlO catalyst, much higher than the counterpart of NiMgAlO catalyst (51.8 $\mu\text{mol/g}$). This tendency is consistent with the CO adsorption result in Fig. S6, confirming the elevated metallic sites of Cu-NiMgAlO catalysts for this reaction.

Ethanol dehydrogenation is the first and rate-determining step for the conversion of ethanol to *n*-butanol [56]. Compared with the NiMgAlO catalyst, the presence of Cu greatly increased the number of metal active sites and enhanced hydrogen transfer activity [57]. As a result, the ethanol dehydrogenation reaction was accelerated, and the ethanol conversion was greatly improved, as shown in Fig. 1. On the other hand, when the Cu loading was overmuch, e.g. the 4%Cu-NiMgAlO catalyst, bulk CuNi alloys were formed, which led to a decrease of the dehydrogenation activity and low ethanol conversions.

Besides the hydrogen transfer reaction, acetaldehyde condensation is the other key elementary step in the ethanol to *n*-butanol process, which is highly dependent on the acidic and basic properties of catalysts. Fig. 9(a) depicted NH_3 -TPD profiles of NiMgAlO and 2%Cu-NiMgAlO catalysts. Similar to the NiMgAlO catalyst, two main peaks at 376 K and 984 K were observed for the 2%Cu-

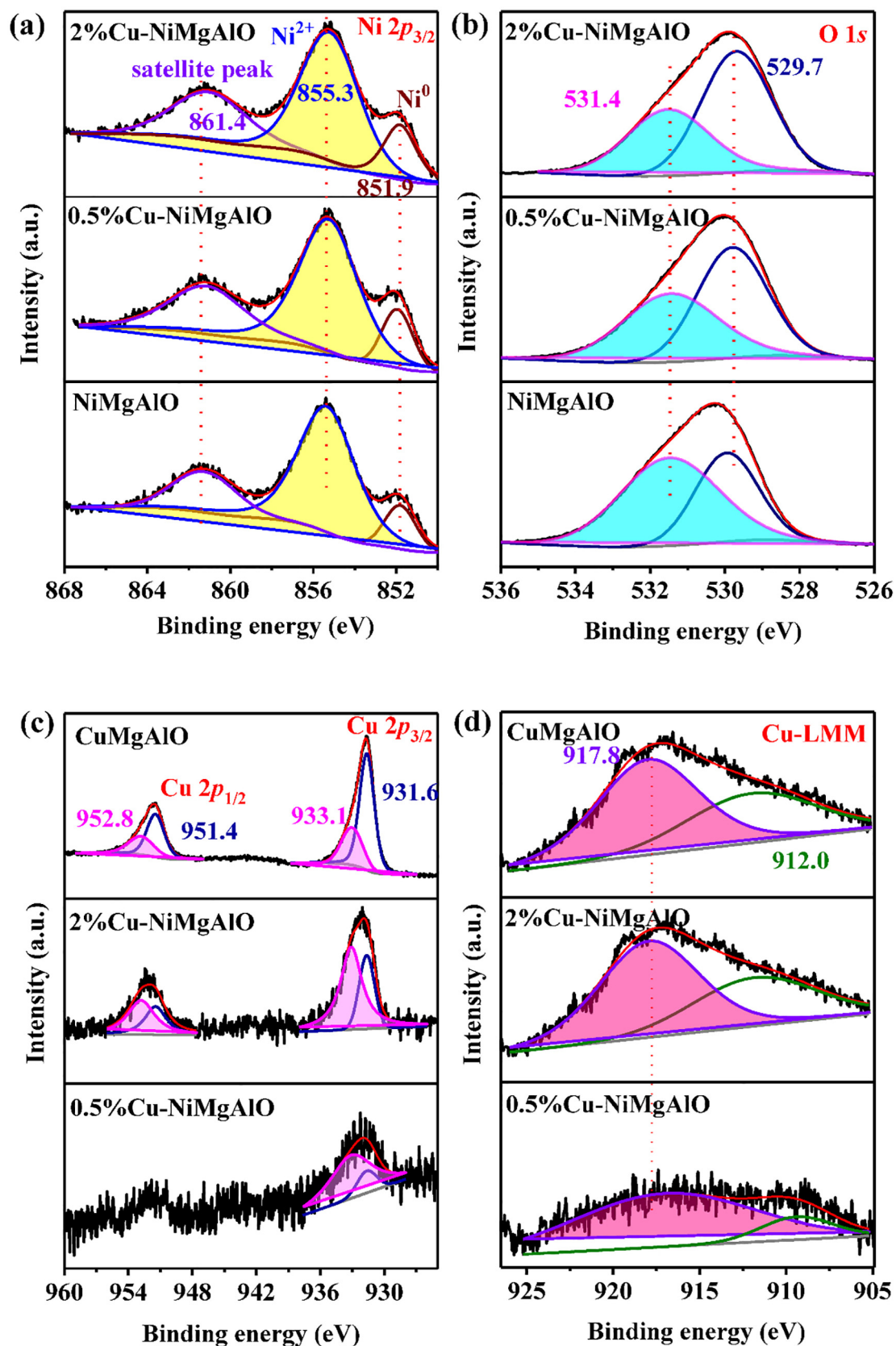


Fig. 6. XPS spectra of (a) Ni 2p_{3/2} and (b) O 1s for NiMgAlO, 0.5%Cu-NiMgAlO, 2%Cu-NiMgAlO catalysts, and (c) Cu 2p and (d) Cu-LMM for 0.5%Cu-NiMgAlO, 2%Cu-NiMgAlO and CuMgAlO catalysts.

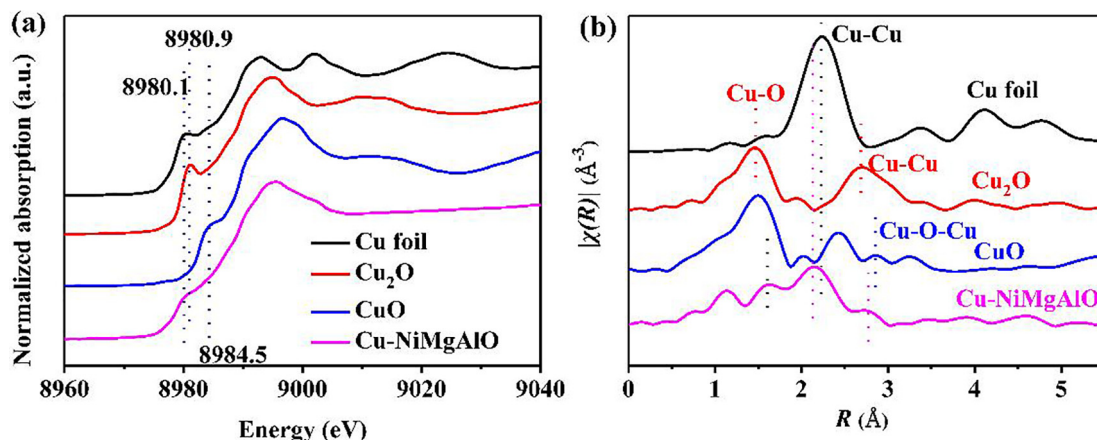


Fig. 7. (a) Normalized XANES spectra and (b) k^3 -weighted Fourier transform spectra from XAFS at the K-edge of Cu foil, Cu₂O, CuO and 2%Cu-NiMgAlO catalysts.

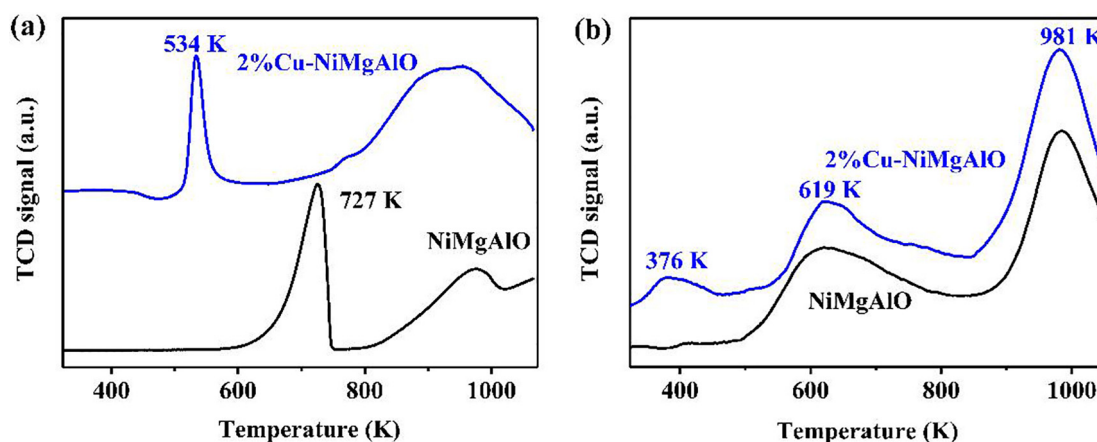


Fig. 8. (a) H₂-TPR and (b) H₂-TPD profiles of NiMgAlO and 2%Cu-NiMgAlO catalysts.

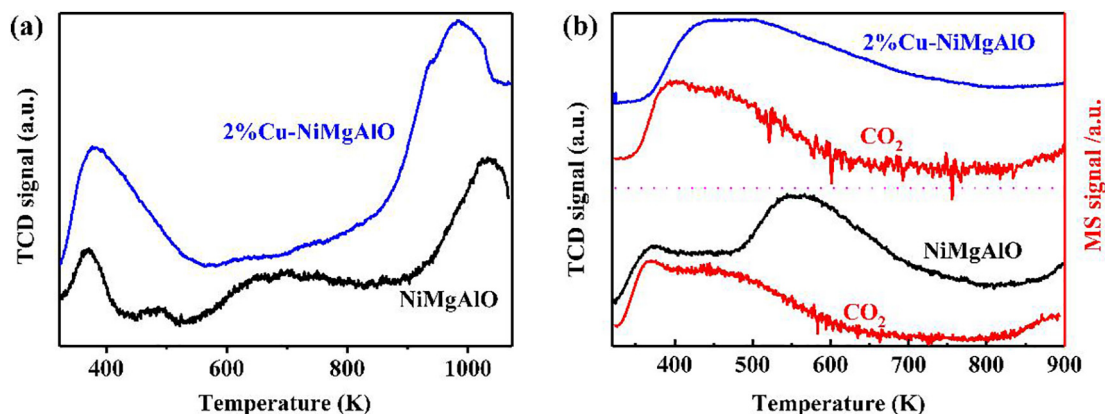


Fig. 9. (a) NH₃-TPD and (b) CO₂-TPD profiles of NiMgAlO and 2%Cu-NiMgAlO catalysts.

NiMgAlO catalyst. According to the MS profile in Fig. S7, the first peak at ca. 376 K should be attributed to the NH₃ desorption, and ascribed to weak acid sites. Compared with the NiMgAlO catalyst, the presence of Cu in the 2%Cu-NiMgAlO catalyst enlarged the acid amount from 371 $\mu\text{mol/g}$ to 609 $\mu\text{mol/g}$. According to pyridine desorption experiments in Fig. S8 and Table S2, only Lewis acid sites were observed and the increased acid sites should be mainly attributed to the medium and strong Lewis acids [58]. Differently, Cu species had negligible influence on total base sites of

NiMgAlO catalysts (56.4 $\mu\text{mol/g}$ for the NiMgAlO catalyst vs 51.5 $\mu\text{mol/g}$ for the 2%Cu-NiMgAlO catalyst), but slightly increased the basicity strength, as evidenced by the increased CO₂ desorption temperature in Fig. 9(b). This result is in good accordance with the elevated lattice oxygen species in Fig. 6(b).

To depress side reactions of ethanol dehydration or decomposition, suitable Lewis acid or base sites were constructed for acetaldehyde condensation in the conversion of ethanol to *n*-butanol [59]. Besides solo acid or base sites, acid-base pairs

demonstrated elevated activity in condensation reactions [60]. Over the NiMgAlO catalyst, weak Lewis acid and strong base sites were detected, which minimized ethanol dehydration reactions but also decreased ethanol conversions [35]. Differently, the Cu doped NiMgAlO catalyst demonstrated increased medium and strong Lewis acid sites with enhanced basic properties, forming abundant acid-base pairs, which promoted ethanol dehydrogenation and condensation reactions, and then elevated ethanol conversion and *n*-butanol selectivity. More important, the presence of Cu species did not draw in Brønsted acid sites, so the possible side reactions of ethanol dehydration or decomposition were inhibited.

3.2.4. Theoretical calculations of acetaldehyde condensation over Mg and Cu-Mg sites

To gain an atomic level understanding of condensation reactions over Cu-NiMgAlO catalysts, the elementary step for acetaldehyde coupling on MgO (111), CuO and Cu₂O doped MgO (111) surfaces were calculated theoretically. As shown in Fig. 10 and Fig. S9, over Mg-O-Mg sites, acetaldehyde molecules were separately adsorbed in linear and bridge manners with an energy barrier of 8.3 kcal/mol [61,62]. Then, two adsorbed acetaldehyde molecules preferably formed a cyclic structure of TS23 with a barrier of 0.2 kcal/mol. After the C-C bond formation, the dehydration reaction of adsorbed acetaldol to adsorbed crotonaldehyde (TS56) was found to have the highest energy barrier of 9.8 kcal/mol. Unlike the higher energy barrier on Cu(2+)-O-Mg sites (14.5 kcal/mol in Fig. S10), the adsorption barrier of acetaldehyde decreased to 5.3 kcal/mol over Cu(+)-O-Mg sites. Simultaneously, the energy barriers for cycling acetaldehyde molecules (TS23) and adsorbed crotonaldehyde formation (TS56) were largely decreased to -12.5 and -6.7 kcal/mol, respectively. In accordance with our previous characterizations, the presence of Cu(+) species created Lewis acid sites of Cu(+)-O-Mg, which newly formed acid-base pairs for promoting the adsorption and condensation of acetaldehyde and then elevated the final product yields.

3.2.5. DRIFTS of NiMgAlO and Cu-NiMgAlO catalysts

In situ DRIFTS experiments were carried out using CO as the probe molecule to revisit the chemical state of Cu species. As shown in Fig. 11(a), negligible CO adsorption was observed for the NiMgAlO catalyst after purging He for 15 min. After introducing Cu species into NiMgAlO, a symmetric peak centered at around 2105 cm⁻¹ appeared, and should be ascribed to the CO adsorption

on Cu⁺ species [63]. Meanwhile, a weak band at 2057 cm⁻¹ assigned to Cu⁰(Ni)-CO was also detected, indicating the partially reduced Cu in Ni on the surface of catalysts. These results are consistent with XPS and XAFS conclusions in Figs. 6 and 7, confirming the co-existing of Cu⁺ and Cu⁰(Ni) on the surface of Cu-NiMgAlO catalysts.

In situ DRIFTS of ethanol adsorption and desorption over different catalysts were then performed. After saturation with ethanol at 298 K (dose ethanol), the catalyst was purged with flowing Ar to remove the gas phase and weakly adsorbed ethanol molecules. Then, the cell temperature was gradually increased from 298 K to 523 K with a heating rate of 10 K/min. The DRIFT spectra obtained after subtraction of the background were shown in Fig. 11(b). Over the NiMgAlO catalyst at 298 K, typical bands at 2973 cm⁻¹, 1103 cm⁻¹ and 1052 cm⁻¹ were observed, and should be assigned to the asymmetric vibration of the CH₃ and C-O stretching of ethanol, respectively [64–66]. As increasing the reaction temperature, bands at 1103 cm⁻¹ and 1052 cm⁻¹ gradually disappeared and distinct bands at 1765 and 1595 cm⁻¹ attributed to conjugate C=C and C=O vibrations were produced, indicating the transformation of ethanol to crotonaldehyde intermediates [67,68], in line with the theoretical calculation results in Fig. 10. Apart from the main reaction pathway for coupling products, some esters, olefins and organic acids were also detected due to the appearance of C-O-C bond at 1076 cm⁻¹ and $\nu(\text{OCO})$ bond at 1595 cm⁻¹ at temperatures >323 K. After doping 2% Cu, a low wavenumber band of 1170 cm⁻¹ was detected due to ethanol molecules coordinated at Lewis acid sites [69]. Meanwhile, the aldehyde bond vibration at 1765 cm⁻¹ disappeared at the whole temperature range, and a new band at 1638 cm⁻¹ assigned to adsorbed C=O of butyraldehyde appeared [70]. Correlating to the different structures between NiMgAlO and Cu-NiMgAlO catalyst, Cu⁺/Cu⁰(Ni) species promoted the transformation of ethanol to aldehydes, and then to butyraldehyde due to the enhanced acid-base pairs and metal sites. As increasing reaction temperatures, the $\nu(\text{C=O})$ band gradually disappeared, indicating the hydrogenation of C=O bonds of butyraldehyde via the adsorbed hydrogen on CuNi alloys [71,72].

3.2.6. Reaction pathways for conversion of ethanol to *n*-butanol over NiMgAlO and Cu-NiMgAlO catalysts

According to the above characterizations and *in situ* experiments, the reaction pathway for conversion of ethanol to *n*-

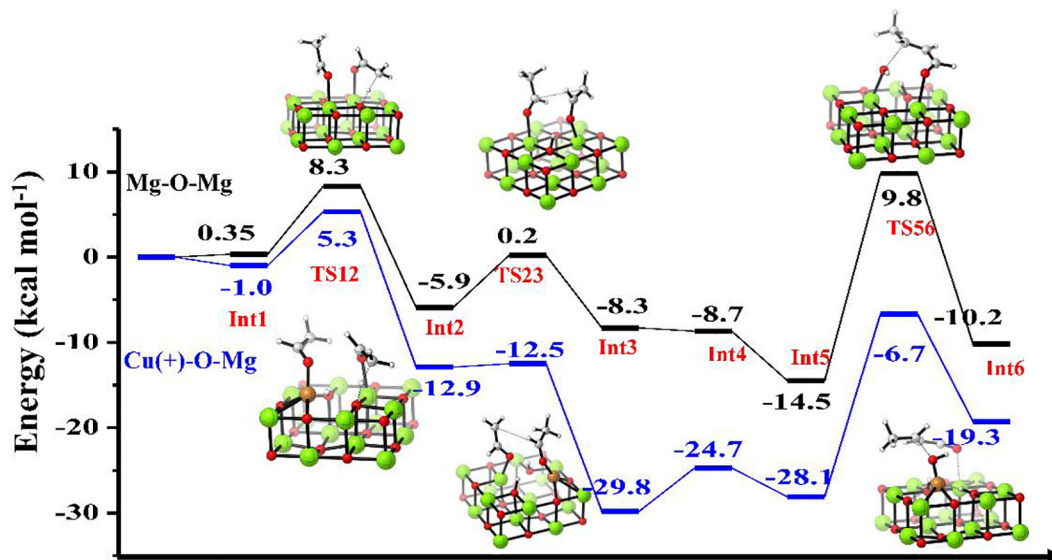


Fig. 10. Gibbs free energy profile for acetaldehyde condensation over Mg-O-Mg and Cu(+)-O-Mg sites.

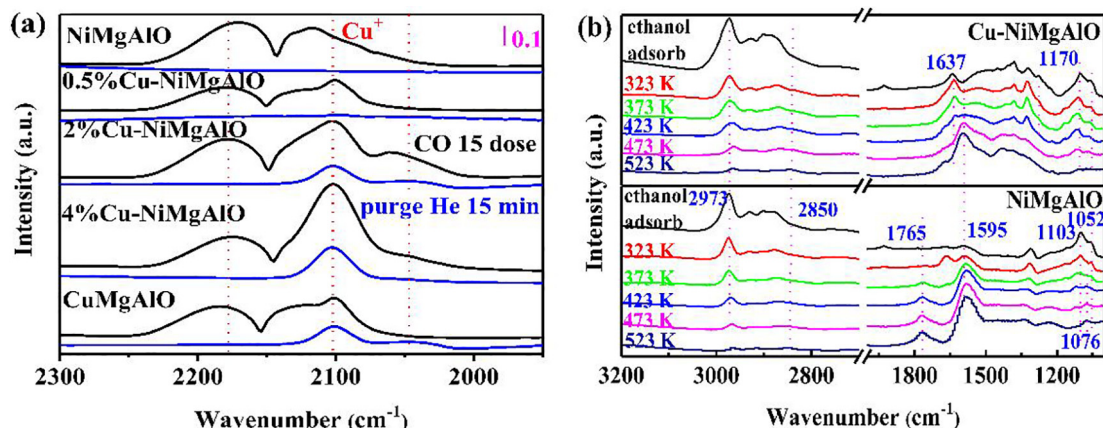
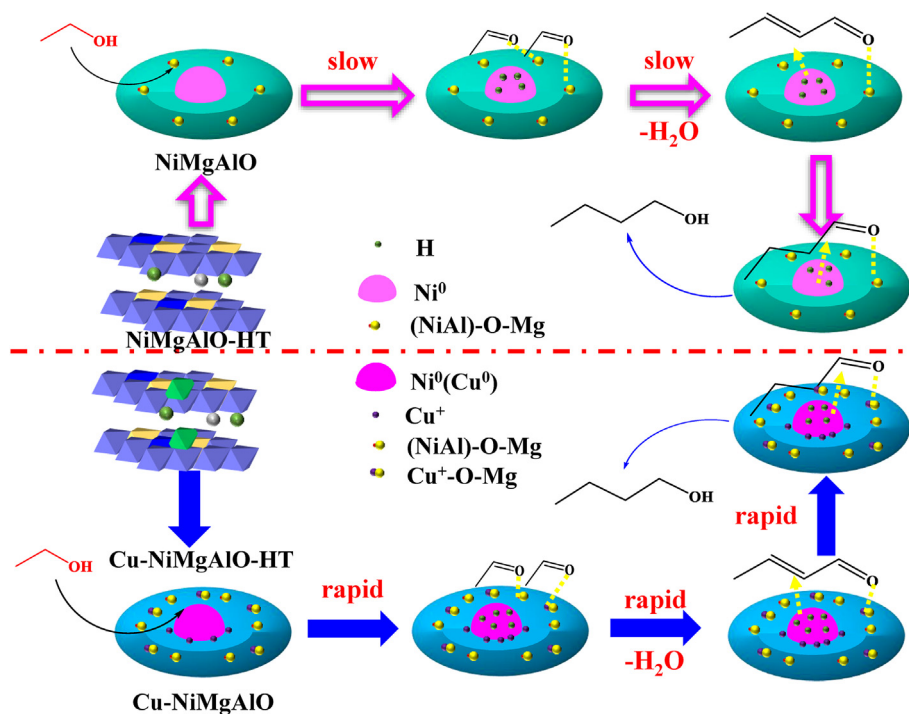


Fig. 11. In situ DRIFTS spectra of (a) CO adsorption and (b) ethanol adsorption and transformation on reduced NiMgAlO and 2%Cu-NiMgAlO catalysts.



Scheme 2. Conversion of ethanol to *n*-butanol over NiMgAlO and Cu-NiMgAlO catalysts.

butanol over NiMgAlO and Cu-NiMgAlO catalysts was proposed. As shown in Scheme 2, over the partially reduced Ni species on NiMgAlO catalyst, ethanol was dehydrogenated into acetaldehyde and hydrogen, which were adsorbed on MgAl oxides and Ni particles, respectively. Then, the produced acetaldehyde molecules were condensed to crotonaldehyde over weak acid and strong base sites, as we previously reported [35]. Finally, the adsorbed crotonaldehyde was hydrogenated to *n*-butanol with quick desorption. During this process, some ester and organic acid species were detected to strongly adsorb on the NiMgAlO catalyst. Differently, the introduction of Cu species formed abundant Cu⁺ species for O–H and C–H bonds activation, and strengthened CuNi metal sites for hydrogen atoms storage, which finally enhanced ethanol conversions. Simultaneously, Cu species in NiMgAlO created large amounts of medium and strong Lewis acid sites, forming abundant acid-base pairs of Cu(+)-O-Mg and lattice oxygen. They demonstrated higher activity in condensation reactions, and then elevated condensed products selectivity [73]. In short, the presence of Cu

species on NiMgAlO enlarged the number of metal sites and created considerable Lewis-base pairs, which promoted ethanol dehydrogenation, crotonaldehyde hydrogenation and condensation reactions, respectively, and resulted in the enhanced ethanol conversion and *n*-butanol selectivity.

4. Conclusions

Different transition metals were screened to dope NiMgAlO catalysts for conversion of ethanol to *n*-butanol. Compared with the benchmark catalyst of NiMgAlO, the presence of Cu species notably promoted ethanol conversion and *n*-butanol selectivity. Over the optimized 2%Cu-NiMgAlO catalyst, ethanol conversion and *n*-butanol selectivity reached 30.0% and 64.2%, respectively, under the reaction conditions of 523 K and a WHSV of 1.69 h^{−1}. More important, this catalyst was used for more than 200 h without any deactivation, demonstrating high potential for applications.

For the optimized 2%Cu-NiMgAlO catalyst, the NiMgAlO hydroxalite structure was maintained to yield plate-like catalysts with enlarged surface areas and an average Ni particle size of 5.4 nm. The presence of Cu species created large amounts of Cu⁺ and CuNi alloys sites, and then increased the hydrogen transfer ability. Moreover, over the 2%Cu-NiMgAlO catalyst, medium and strong Lewis acid sites, and strengthened base sites were created, and significantly accelerated condensation reactions. Along with the strong interaction between CuNi and CuNiMgAl oxides with close ionic radius and lattice matching, these metal sites and acid-base pairs were coupled to reserve the high ethanol conversion, *n*-butanol selectivity and catalyst stability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jechem.2022.04.049>.

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